Complete Guide: Android Development in Termux

This documentation covers how to build, compile, and deploy a fully functional Android app (StudySpark) using only an Android phone, Termux, and GitHub Actions.

Prerequisites

1. **Termux** installed (from F-Droid).
2. A **GitHub Account** (free).
3. A **Google AI Studio Account** (for the Gemini API keys).

Step 1: Termux & Ubuntu Setup

1. Update Termux and install Ubuntu:
pkg update && pkg upgrade -y
pkg install proot-distro -y
proot-distro install ubuntu

2. Log into Ubuntu:
proot-distro login ubuntu

3. Install development tools:
apt update && apt upgrade -y
apt install openjdk-17-jdk git wget curl unzip nano -y

Step 2: Project Initialization

1. Create the directory structure:
mkdir -p ~/StudySpark/app/src/main/java/com/studyspark/app
mkdir -p ~/StudySpark/app/src/main/res/values
mkdir -p ~/StudySpark/.github/workflows
cd ~/StudySpark

2. Create the Gradle Wrapper:
mkdir -p gradle/wrapper
curl -L -o gradle/wrapper/gradle-wrapper.jar
https://github.com/gradle/gradle/raw/v8.5.0/gradle/wrapper/gradle-wrapper.jar
curl -L -o gradlew https://github.com/gradle/gradle/raw/v8.5.0/gradlew
chmod +x gradlew

Step 3: Core Plugins & Dependencies

1. Project Level build.gradle.kts:
plugins {
id("com.android.application") version "8.2.2" apply false

id("org.jetbrains.kotlin.android") version "1.9.22" apply false
id("com.google.devtools.ksp") version "1.9.22-1.0.16" apply false
id("org.jetbrains.kotlin.plugin.serialization") version "1.9.22" apply false
}

2. App Level Dependencies:
- Compose: androidx.compose:compose-bom:2024.02.00
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Step 4: CI/CD Setup (GitHub Actions)

- Create .github/workflows/android-build.yml
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Step 5: Advanced Configurations

- **FileProvider:** Added to AndroidManifest for saving/sharing PDFs.
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How to Deploy on a New Phone

1. Setup Termux, Ubuntu, and Java.
2. git clone https://github.com/Prudent66/StudySpark.git
3. Create GitHub Secrets for API keys.
4. Edit code with nano, git push, and download APK from GitHub Actions.